

Funciones Hiperbólicas:

$$1. \sinh x = \frac{e^x - e^{-x}}{2} \quad 2. \cosh x = \frac{e^x + e^{-x}}{2}$$

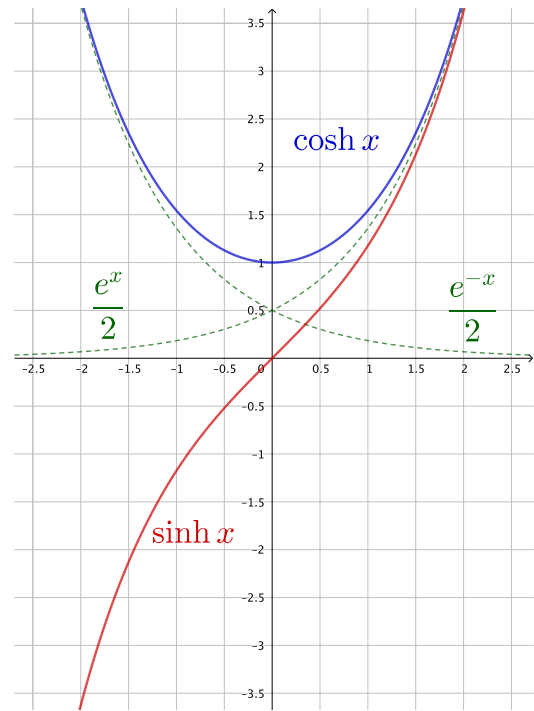
Funciones obtenidas:

$$1. \tanh x = \frac{\sinh x}{\cosh x} = \frac{e^x - e^{-x}}{e^x + e^{-x}}$$

$$2. \coth x = \frac{\cosh x}{\sinh x} = \frac{e^x + e^{-x}}{e^x - e^{-x}}, \quad x \neq 0$$

$$3. \operatorname{sech} x = \frac{1}{\cosh x} = \frac{2}{e^x + e^{-x}}$$

$$4. \operatorname{csch} x = \frac{1}{\sinh x} = \frac{2}{e^x - e^{-x}}, \quad x \neq 0$$



Identidades Hiperbólicas fundamentales:

$$1. \cosh^2 x - \sinh^2 x = 1 \quad 2. \operatorname{sech}^2 x = 1 - \tanh^2 x \quad 3. \operatorname{csch}^2 x = \coth^2 x - 1$$

Paridad:

$$1. \sinh(-x) = -\sinh x \quad 2. \cosh(-x) = \cosh x$$

Más identidades:

$$1. \sinh(x + y) = \sinh x \cosh y + \sinh y \cosh x$$

$$2. \cosh(x + y) = \cosh x \cosh y + \sinh x \sinh y$$

$$3. \sinh(2x) = 2 \sinh x \cosh x$$

$$4. \cosh(2x) = \cosh^2 x + \sinh^2 x$$

$$5. \sinh^2\left(\frac{x}{2}\right) = \frac{\cosh(x) - 1}{2}$$

$$6. \cosh^2\left(\frac{x}{2}\right) = \frac{\cosh(x) + 1}{2}$$

Funciones hiperbólicas inversas

Definiciones:

1. $\sinh^{-1} x = \ln \left(x + \sqrt{x^2 + 1} \right), \forall x \in \mathbb{R}.$
2. $\cosh^{-1} x = \ln \left(x + \sqrt{x^2 - 1} \right), \forall [1, +\infty[.$
3. $\tanh^{-1} x = \frac{1}{2} \ln \left(\frac{1+x}{1-x} \right), \forall x \in]-1, 1[.$
4. $\coth^{-1} x = \frac{1}{2} \ln \left(\frac{x+1}{x-1} \right), \forall x \in]-\infty, -1[\cup]1, +\infty[.$
5. $\operatorname{sech}^{-1} x = \ln \left(\frac{1 + \sqrt{1 - x^2}}{x} \right), \forall x \in]0, 1[.$
6. $\operatorname{csch}^{-1} x = \ln \left(\frac{1}{x} + \frac{\sqrt{1 + x^2}}{|x|} \right), \forall x \in \mathbb{R} \setminus \{0\}.$

Identidades:

1. $\coth^{-1} x = \tanh^{-1} \left(\frac{1}{x} \right)$
2. $\operatorname{sech}^{-1} x = \cosh^{-1} \left(\frac{1}{x} \right)$
3. $\operatorname{csch}^{-1} x = \sinh^{-1} \left(\frac{1}{x} \right)$

Derivadas e integrales de funciones hiperbólicas

Derivadas básicas:

- | | |
|--|---|
| 1. $\frac{d}{dx}(\sinh x) = \cosh x$ | 4. $\frac{d}{dx}(\coth x) = -\operatorname{csch}^2 x$ |
| 2. $\frac{d}{dx}(\cosh x) = \sinh x$ | 5. $\frac{d}{dx}(\operatorname{sech} x) = -\operatorname{sech} x \tanh x$ |
| 3. $\frac{d}{dx}(\tanh x) = \operatorname{sech}^2 x$ | 6. $\frac{d}{dx}(\operatorname{csch} x) = -\operatorname{csch} x \coth x$ |

Ejemplo 1.

$$\frac{d}{dx}(\sinh x) = \frac{d}{dx} \left(\frac{e^x - e^{-x}}{2} \right) = \frac{e^2 + e^{-x}}{2} = \cosh x$$

Derivadas de las funciones hiperbólicas inversas:

$$1. \frac{d}{dx} (\sinh^{-1} x) = \frac{1}{\sqrt{1+x^2}}$$

$$4. \frac{d}{dx} (\coth^{-1} x) = \frac{1}{1-x^2}$$

$$2. \frac{d}{dx} (\cosh^{-1} x) = \frac{1}{\sqrt{x^2-1}}$$

$$5. \frac{d}{dx} (\operatorname{sech}^{-1} x) = -\frac{1}{x\sqrt{1-x^2}}$$

$$3. \frac{d}{dx} (\tanh^{-1} x) = \frac{1}{1-x^2}$$

$$6. \frac{d}{dx} (\operatorname{csch}^{-1} x) = -\frac{1}{|x|\sqrt{1+x^2}}$$

Integrales inmediatas:

$$1. \int \sinh x \, dx = \cosh x + C$$

$$2. \int \cosh x \, dx = \sinh x + C$$

$$3. \int \operatorname{sech}^2 x \, dx = \tanh x + C$$

$$4. \int \operatorname{csch}^2 x \, dx = -\coth x + C$$

$$5. \int \operatorname{sech} x \tanh x \, dx = -\operatorname{sech} x + C$$

$$6. \int \operatorname{csch} x \coth x \, dx = -\operatorname{csch} x + C$$

$$7. \int \frac{dx}{\sqrt{x^2+1}} = \sinh^{-1} x + C$$

$$8. \int \frac{dx}{\sqrt{x^2-1}} = \cosh^{-1} x + C$$

$$9. \int \frac{dx}{\sqrt{1-x^2}} = \tanh^{-1} x + C, \forall x \in]-1, 1[.$$

$$10. \int \frac{dx}{\sqrt{1-x^2}} = \coth^{-1} x + C, \forall x \in]-\infty, -1[\cup]1, +\infty[.$$

$$11. \int \frac{dx}{\sqrt{1-x^2}} = \frac{1}{2} \ln \left| \frac{1+x}{1-x} \right| + C.$$

$$12. \int \frac{dx}{x\sqrt{1-x^2}} = -\operatorname{sech}^{-1}|x| + C.$$

$$13. \int \frac{dx}{x\sqrt{1+x^2}} = -\operatorname{csch}^{-1}|x| + C.$$

Método de Sustitución Hiperbólica:

Ejemplo 2. Calcule las siguientes integrales indefinidas

1. $\int \frac{dx}{\sqrt{4x^2 + 1}}$

Solución: Aplicamos el cambio de variables $u = 2x$, $dx = \frac{1}{2} du$

$$\int \frac{dx}{\sqrt{4x^2 + 1}} = \frac{1}{2} \int \frac{du}{\sqrt{u^2 + 1}} = \frac{1}{2} \sinh^{-1}(2x) + C.$$

2. $\int \frac{dx}{x\sqrt{4 + x^2}}$

Solución:

Ejemplo 3. Calcule las siguientes integrales de Riemann

1. $\int_0^{1/2} \frac{dx}{1 - x^2}$

Solución:

$$\int_0^{1/2} \frac{dx}{1 - x^2} = \tanh^{-1} x \Big|_0^{1/2} = \frac{1}{2} \ln \left| \frac{1+x}{1-x} \right| \Big|_0^{1/2} = \frac{1}{2} \ln 3$$

2.

Solución:

$$\int_2^5 \frac{dx}{1 - x^2} = \coth^{-1} x \Big|_2^5 = \frac{1}{2} \ln \left| \frac{1+x}{1-x} \right| \Big|_2^5 = -\frac{1}{2} \ln 2$$

Ejercicios: Revisar los Problemas 6.9 desde la pag 506 del libro “Cálculo con trascendentes tempranas” de Edwards y Penney